always true: rather, it is usually not the case, for generic graphs.

To put it differently, the extra information here doesn't come from the graph, but from domain knowledge.

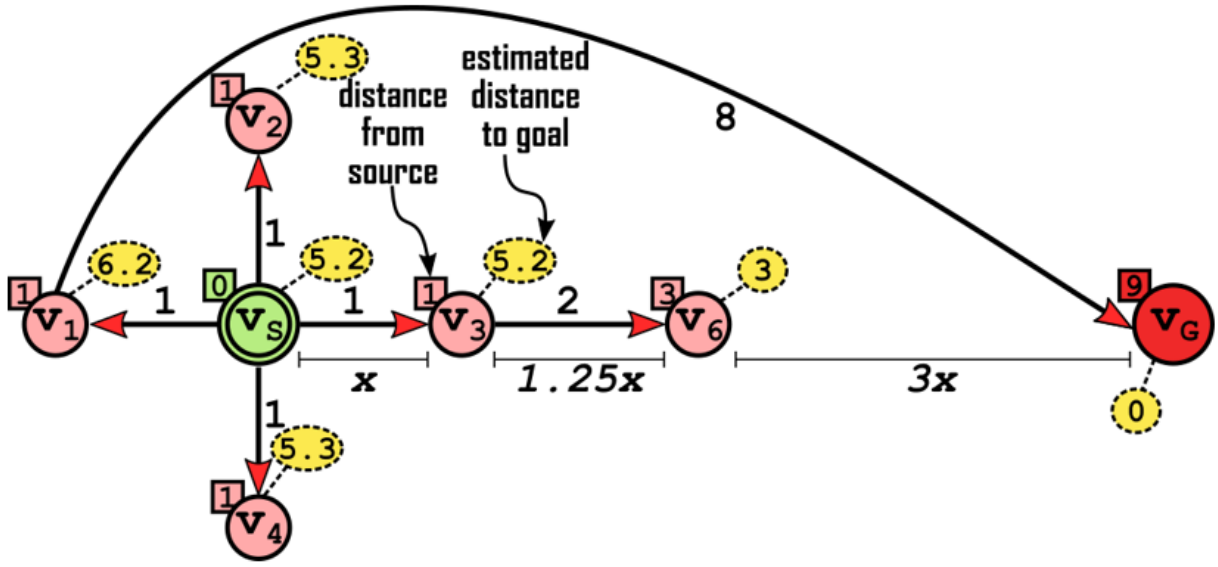


Figure 14.20 An edge case where A* certainly can't out-perform Dijkstra: both algorithms will visit all vertices (A* always in the same order, Dijkstra in a partially random order), before eventually travelling the edge from v_1 to v_G.

The good news is somehow limited though, because there is no *a-priori* guarantee that A* will perform better than Dijkstra: on the contrary, it's easy to craft an example where A* will always do as badly as Dijkstra. Check out figure 14.20 to get an idea how we can tweak our previous example to fool A*! The key, here and always, is the quality of the extra information captured by the heuristic function: the more reliable and closer to real distance is the estimate, the better A* performs.